

STUDY QUESTIONS

Industrial Engineering Problems

ANSWER KEY

1. **A Local Bakery:**

a) Critical ratio:

$$\frac{20}{27} \approx 0.741$$

b) Sorted data:

38, 41, 42, 43, 44, 45, 46, 46, 47, 47,
48, 48, 49, 49, 50, 50, 51, 52, 52, 53,
53, 54, 55, 55, 56, 57, 58, 59, 60, 61

Position corresponding to 0.741: $30 \times 0.741 = 22.23 \approx 22\text{nd position}$
The 22nd smallest value is 54.

c) **54 loaves** per day.

2. **A flower shop:**

a) Critical ratio:

$$\frac{2}{3} \approx 0.667$$

b) Sorted data:

58, 59, 60, 61, 62, 63, 63, 64, 64, 65,
65, 66, 66, 67, 67, 68, 68, 69, 69, 69,
70, 70, 71, 71, 72, 72, 73, 74, 75, 76

Position corresponding to 0.667: $30 \times 0.667 = 20\text{th position}$
The 20th smallest value is 69.

c) **69 bouquets** per day.

3. **An electronics retailer:**

a) Critical ratio:

$$F(x^*) = \frac{p - c}{p - s} = \frac{1,200 - 800}{1,200 - 500} = \frac{400}{700} = \frac{4}{7} \approx 0.571$$

b) Optimal order quantity:

$$= 214.29 \approx \mathbf{214 \text{ units}}$$

c) New critical ratio:

$$\frac{3}{7} \approx 0.429$$

New optimal quantity:

$$185.71 \approx \mathbf{186 \text{ units}}$$

DECREASES

4. Emergency Service Coverage:

Iteration 1:

- Uncovered elements: $U' = \{1, 2, 3, 4, 5, 6, 7, 8\}$
- Efficiency calculations:
 - $S_1: \frac{3}{12,000} = 0.00025$
 - $S_2: \frac{3}{15,000} = 0.0002$
 - $S_3: \frac{4}{20,000} = 0.0002$
 - $S_4: \frac{3}{10,000} = 0.0003 \leftarrow \textbf{Highest}$
 - $S_5: \frac{3}{18,000} = 0.000167$
- **Select** S_4 (covers $\{4, 7, 8\}$)
- Covered elements: $\{4, 7, 8\}$
- Remaining uncovered: $U' = \{1, 2, 3, 5, 6\}$

Iteration 2:

- Uncovered elements: $U' = \{1, 2, 3, 5, 6\}$
- Efficiency calculations:
 - $S_1: \frac{3}{12,000} = 0.00025$
 - $S_2: \frac{2}{15,000} = 0.000133$ (covers 2 and 5 from uncovered)
 - $S_3: \frac{3}{20,000} = 0.00015$ (covers 3, 5, and 6 from uncovered)
 - $S_5: \frac{2}{18,000} = 0.000111$ (covers 1 and 6 from uncovered)
- **Select** S_1 (covers $\{1, 2, 3\}$)
- Covered elements: $\{1, 2, 3, 4, 7, 8\}$
- Remaining uncovered: $U' = \{5, 6\}$

Iteration 3:

- Uncovered elements: $U' = \{5, 6\}$
- Efficiency calculations:
 - $S_2: \frac{1}{15,000} = 0.0000667$ (only covers 5 from uncovered)
 - $S_3: \frac{2}{20,000} = 0.0001 \leftarrow \textbf{Highest}$ (covers 5 and 6)
 - $S_5: \frac{1}{18,000} = 0.0000556$ (only covers 6 from uncovered)
- **Select** S_3 (covers $\{3, 5, 6, 7\}$, but only 5 and 6 are new)
- All elements now covered: $U' = \emptyset$

Final Solution:

- Selected sets: S_4, S_1, S_3
- Total cost: $10,000 + 12,000 + 20,000 = \textbf{42,000 TL}$
- Coverage: All 8 neighborhoods are covered

5. Wi-Fi Router Placement:

Iteration 1:

- Uncovered elements: $U' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- Efficiency calculations:
 - $S_1: \frac{4}{8,000} = 0.0005$
 - $S_2: \frac{5}{10,000} = 0.0005$
 - $S_3: \frac{4}{9,000} = 0.000444$

- $S_4: \frac{5}{12,000} = 0.000417$
- $S_5: \frac{3}{6,000} = 0.0005$
- $S_6: \frac{5}{11,000} = 0.000455$
- Three sets tie with efficiency 0.0005. We can select any one. Let's select S_2 (covers $\{3, 4, 5, 6, 7\}$)
- Covered elements: $\{3, 4, 5, 6, 7\}$
- Remaining uncovered: $U' = \{1, 2, 8, 9, 10\}$

Iteration 2:

- Uncovered elements: $U' = \{1, 2, 8, 9, 10\}$
- Efficiency calculations:
 - $S_1: \frac{2}{8,000} = 0.00025$ (covers 1 and 2)
 - $S_3: \frac{2}{9,000} = 0.000222$ (covers 8 and 9)
 - $S_4: \frac{5}{12,000} = 0.000417 \leftarrow$ **Highest** (covers all 5 remaining)
 - $S_5: \frac{2}{6,000} = 0.000333$ (covers 9 and 10)
 - $S_6: \frac{3}{11,000} = 0.000273$ (covers 2, 8, and 10)
- **Select** S_4 (covers $\{1, 2, 8, 9, 10\}$)
- All elements now covered: $U' = \emptyset$

Final Solution:

- Selected sets: S_2, S_4
- Total cost: $10,000 + 12,000 = \mathbf{22,000 \text{ TL}}$
- Coverage: All 10 buildings are covered

6. Security Camera Installation:

Iteration 1:

- Uncovered elements: $U' = \{A, B, C, D, E, F\}$
- Efficiency calculations:
 - $S_1: \frac{3}{6,000} = 0.0005$
 - $S_2: \frac{3}{8,000} = 0.000375$
 - $S_3: \frac{3}{7,000} = 0.000429$
 - $S_4: \frac{3}{9,000} = 0.000333$
 - $S_5: \frac{2}{5,000} = 0.0004$
- **Select** S_1 (highest efficiency, covers $\{A, B, C\}$)
- Covered elements: $\{A, B, C\}$
- Remaining uncovered: $U' = \{D, E, F\}$

Iteration 2:

- Uncovered elements: $U' = \{D, E, F\}$
- Efficiency calculations:
 - $S_2: \frac{2}{8,000} = 0.00025$ (covers D and E)
 - $S_3: \frac{1}{7,000} = 0.000143$ (only covers D)
 - $S_4: \frac{3}{9,000} = 0.000333 \leftarrow$ **Highest** (covers all 3 remaining)
 - $S_5: \frac{1}{5,000} = 0.0002$ (only covers F)
- **Select** S_4 (covers $\{D, E, F\}$)
- All elements now covered: $U' = \emptyset$

Final Solution:

- Selected sets: S_1, S_4
- Total cost: $6,000 + 9,000 = \mathbf{15,000 \text{ TL}}$
- Coverage: All 6 areas (A, B, C, D, E, F) are covered

7. **A delivery company:**
Completed DP Table:

Capacity (w)	0	1	2	3	4	5	6	7	8
Item-null	0	0	0	0	0	0	0	0	0
Item A (w=2)	0	0	10	10	10	10	10	10	10
Item B (w=3)	0	0	10	15	15	25	25	25	25
Item C (w=4)	0	0	10	15	20	25	30	35	35
Item D (w=5)	0	0	10	15	20	25	30	35	40

Sample Calculations:**Cell $V(1, 2)$ (Item A, capacity 2):**

- $w_A = 2 \leq 2$, so we can choose item A
- $V(1, 2) = \max\{V(0, 2), p_A + V(0, 2 - 2)\}$
- $V(1, 2) = \max\{0, 10 + 0\} = 10$

Cell $V(2, 5)$ (Item B, capacity 5):

- $w_B = 3 \leq 5$, so we can choose item B
- $V(2, 5) = \max\{V(1, 5), p_B + V(1, 5 - 3)\}$
- $V(2, 5) = \max\{10, 15 + 10\} = \max\{10, 25\} = 25$

Cell $V(4, 8)$ (Item D, capacity 8):

- $w_D = 5 \leq 8$, so we can choose item D
- $V(4, 8) = \max\{V(3, 8), p_D + V(3, 8 - 5)\}$
- $V(4, 8) = \max\{35, 25 + 15\} = \max\{35, 40\} = 40$

Answers:

- See completed table above
- Maximum value = **40 TL**
- Packages B and D should be selected**
 - Package B: weight 3, value 15
 - Package D: weight 5, value 25
 - Total weight: 8 kg (exactly at capacity)
 - Total value: 40 TL

8. **A traveler:****a) Calculate $V(2, 2)$:**

Item Y has weight $w_Y = 2$, so $w_Y = 2 \leq 2$. We can consider taking item Y.

$$V(2, 2) = \max\{V(1, 2), p_Y + V(1, 2 - 2)\}$$

$$V(2, 2) = \max\{6, 10 + 0\} = \max\{6, 10\} = \mathbf{10}$$

b) Calculate $V(2, 4)$:

Item Y has weight $w_Y = 2$, so $w_Y = 2 \leq 4$. We can consider taking item Y.

$$V(2, 4) = \max\{V(1, 4), p_Y + V(1, 4 - 2)\}$$

$$V(2, 4) = \max\{6, 10 + 6\} = \max\{6, 16\} = \mathbf{16}$$

c) Calculate $V(3, 3)$:

Item Z has weight $w_Z = 3$, so $w_Z = 3 \leq 3$. We can consider taking item Z.

$$V(3, 3) = \max\{V(2, 3), p_Z + V(2, 3 - 3)\}$$

$$V(3, 3) = \max\{16, 12 + 0\} = \max\{16, 12\} = \mathbf{16}$$

(It's better to take items X and Y for total value 16 rather than just item Z for value 12)

d) Calculate $V(3, 5)$:

$$V(3, 5) = \max\{V(2, 5), p_Z + V(2, 5 - 3)\}$$

First, we need $V(2, 5)$:

$$V(2, 5) = \max\{V(1, 5), p_Y + V(1, 5 - 2)\} = \max\{6, 10 + 6\} = 16$$

Now:

$$V(3, 5) = \max\{16, 12 + 10\} = \max\{16, 22\} = \mathbf{22}$$

Complete Table:

Capacity (w)	0	1	2	3	4	5
Item-null	0	0	0	0	0	0
Item X (w=1)	0	6	6	6	6	6
Item Y (w=2)	0	6	10	16	16	16
Item Z (w=3)	0	6	10	16	18	22

9. Investment Decision Scenarios:

Scenario A:

Answer: No, this investment will NOT reduce the overall project completion time.

Scenario B:

Answer: The constraint is NOT tight. Task M is likely not on the critical path (or N is not the critical successor on the critical path). Reducing task M's duration will NOT affect the overall project completion time.

Scenario C:

Answer: No, from a project completion time perspective, the company should NOT allocate budget to speed up the marketing task.

Scenario D:

Answer: Option 2 will successfully reduce the project completion time by 3 days. Option 1 will NOT successfully reduce the project completion time.

10. Multiple Investment Scenarios:

a) The company should consider **Proposals 1 and 3** (Design and Safety Testing).

Proposals that will achieve the goal:

- **Proposal 1 (Design):** Will reduce project time by 2 months

- **Proposal 3 (Safety Testing):** Will reduce project time by 1 month

Proposals that will NOT achieve the goal:

- **Proposal 2 (Parts Procurement):** Will NOT reduce project time - not on critical path
- **Proposal 4 (Market Research):** Will NOT reduce project time - not on critical path

b) No, the CFO's reasoning is **incorrect**.

c) The new project completion time will be **15 months** (reduced by 3 months total).

- Design by 2 months
- Safety Testing by 1 month

The total reduction in the critical path is $2 + 1 = 3$ months. Since both reductions are on the same critical path and there are no multiple critical paths mentioned, the full benefit of both reductions is realized. The original 18 months becomes $18 - 3 = 15$ months.

d) The suggestion could be correct if **Parts Procurement had only 3 days or less of slack initially**.